

NOURHENE AYADI

DevOps, Automation & Cloud Computing Engineer

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TUNIS, TUNISIA

PROFESSIONAL SUMMARY

- Entry-level AI & Cloud/DevOps Engineer with hands-on experience designing and deploying production-grade systems — from a Retrieval-Augmented Generation (RAG) chatbot and an adaptive AI evaluation platform built for a real external client, to containerized CI/CD pipelines on Kubernetes, AWS, and Azure.
- Comfortable owning a project end-to-end: architecture, backend development (Python/FastAPI), infrastructure-as-code (Terraform, Docker, Kubernetes), and monitoring/observability (Prometheus, Grafana).
- Engineering degree in Industrial Computing and Automation (ENIG, 2022–2026) with certifications across AWS, Azure, and Kubernetes. A fast learner who moves quickly from new tools to working systems, with strong communication skills built through direct client-facing project delivery.

CORE QUALIFICATIONS

- Approximately one year of combined hands-on experience through technical internships, client delivery, and applied IT engineering projects.
- Engineering degree in Industrial Computing and Automation with practical exposure to software, cloud, AI, networking, monitoring, and embedded systems.
- Experience deploying secure, containerized applications and automated CI/CD pipelines across AWS, Azure, Kubernetes, and Linux environments.
- Strong troubleshooting, documentation, requirements analysis, stakeholder communication, and end-to-end solution ownership.

SOFT SKILLS

Communication • Problem-Solving • Adaptability • Time Management • Critical Thinking • Work Under Pressure

COMPLETED TRAINING & CREDENTIALS

- AWS Cloud Quest: Cloud Practitioner.
- Microsoft Azure Fundamentals.
- IBM Cloud Computing Fundamentals.
- Kode-Kloud training: Kubernetes, Terraform, and Linux challenges.
- Linux Essentials, Great Learning.
- Problem Solving, Hacker-Rank.

PROFESSIONAL EXPERIENCE

Sep 2025 – Dec 2025 **DevOps Intern & Full-Stack Application Deployment** — VEARDIA

- Containerized and deployed a full-stack production web app (React/Nginx, Laravel/PHP-FPM, MySQL) with Docker Compose for consistent environments.
- Built a GitLab CI/CD pipeline with automated security scanning (Trivy) and deployment stages.
- Implemented full observability with Prometheus, Grafana, Loki, Promtail, and Alertmanager; managed containers via Portainer.

Jul 2024 to Aug 2024 **Engineer Intern & Industrial Computing and Automation** — Center for Research in Microelectronics and Nanotechnology (CRMN)

- Designed and built a partial-discharge detector, including PCB design (KiCad) and integration with an STM32 microcontroller.
- Built a real-time LabVIEW acquisition and analysis interface for sensor data (filtering, sorting, comparison).

March 2024 to June 2024 **GUI Development Intern (C++)** — CodeAlpha

- Built an interactive Qt Widgets interface for a real-time digital guessing game with live feedback.

GRADUATION PROJECT (PFE)

- Hybrid RAG chatbot: combined ChromaDB semantic search with BM25 keyword retrieval via Reciprocal Rank Fusion, then cross-encoder reranking (ms-marco-MiniLM-L-6-v2) and query expansion for precision; served by Llama 3.3-70B (Groq API) with a local Ollama fallback for resilience.
- Speech-based corrective-feedback robot: Whisper Large V3 for speech recognition with phonetic prompt injection; Llama 3.3-70B generating structured, Pydantic-validated pedagogical feedback; gTTS for spoken responses — full voice-in, voice-out correction loop.
- Adaptive evaluation engine: Gemini 2.0 Flash with Google Search Grounding for curriculum-aligned question generation, Llama 3.3-70B for multi-dimensional scoring, producing personalized, per-learner pedagogical reports.
- Owned the system end-to-end: architecture, prompt engineering, backend (FastAPI), UI, and performance evaluation (latency and retrieval-quality benchmarking) for a real external client.

Conversational AI Platform for Institut Molière, ACTIA Technologies (client: Institut Molière)

Designed and built a three-part generative-AI platform for language learning, defended 26/06/2026.

Stack: Python, FastAPI, ChromaDB, BM25/Whoosh, Sentence-Transformers, Llama 3.3-70B (Groq), Ollama, Gemini 2.0 Flash, Whisper Large V3, Pydantic, gTTS

EDUCATION

2022-2026 Gabès, Tunisia (ENIG)	Engineering Degree, Industrial Computing and Automation — National Engineering School of
2019 – 2022 Sousse, Tunisia (ESSTHS)	Preparatory Cycle, Mathematics-Physics — Higher School of Science and Technology, Hammam

TECHNICAL PROJECTS

End-to-End CI/CD Pipeline for a Football Platform — Personal Project, 2024 to 2025

- Built an end-to-end CI/CD pipeline for a multi-service football platform (REST API, Kafka, MongoDB), using Jenkins to automate build, semantic versioning, and deployment.
- Deployed the platform on a Helm-orchestrated K3s cluster with Kubernetes Secrets and ConfigMaps, plus automated health checks for reliable rollouts.
- Managed project tracking, sprint planning, and technical documentation using Jira boards and Confluence, applying an agile workflow end-to-end.

NLP Intent-Detection Chatbot — Academic Project, 2025

- Fine-tuned DistilBERT on the SNIPS dataset for intent classification; documented in a full academic report.

Two-Tier AWS Deployment & AKS Microservice — Personal Projects, 2025 to 2026

- Provisioned AWS infrastructure (VPC, EC2 autoscaling, RDS) with reusable Terraform modules and Ansible playbooks; separately deployed a containerized microservice on Azure AKS with an Azure DevOps CI/CD pipeline.

LecturBot — French Reading-Assistant Robot — Personal Project

- Built a bilingual (EN/AR) speech-correction web app using local models (Ollama, faster-whisper) designed to run on hardware-constrained devices (no GPU, 8GB RAM).

LANGUAGES

Arabic (native) | French (Excellent) | English (V. Good)